

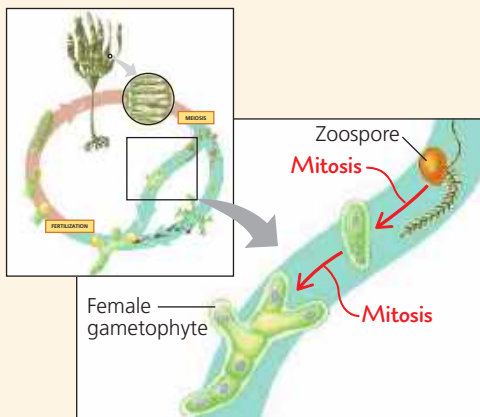
28 Protists

KEY CONCEPTS

- 28.1** Most eukaryotes are single-celled organisms *p. 594*
- 28.2** Excavates include protists with modified mitochondria and protists with unique flagella *p. 597*
- 28.3** SAR is a highly diverse group of protists defined by DNA similarities *p. 601*
- 28.4** Red algae and green algae are the closest relatives of plants *p. 609*
- 28.5** Unikonts include protists that are closely related to fungi and animals *p. 611*
- 28.6** Protists play key roles in ecological communities *p. 614*

Study Tip

Label a drawing: Protist life cycles vary greatly. To make sense of this variation, it can be helpful to identify when mitosis occurs in different life cycles, as shown here for a portion of Figure 28.14. Identify the stages at which mitosis occurs in Figures 28.14, 28.18, 28.25, and 28.28.



Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 28
- Animation: Life Cycle of a Malaria Parasite

For Instructors to Assign (in Item Library)

- Scientific Skills Exercise: Interpreting Comparisons of Genetic Sequences
- Misconception Questions 136–140

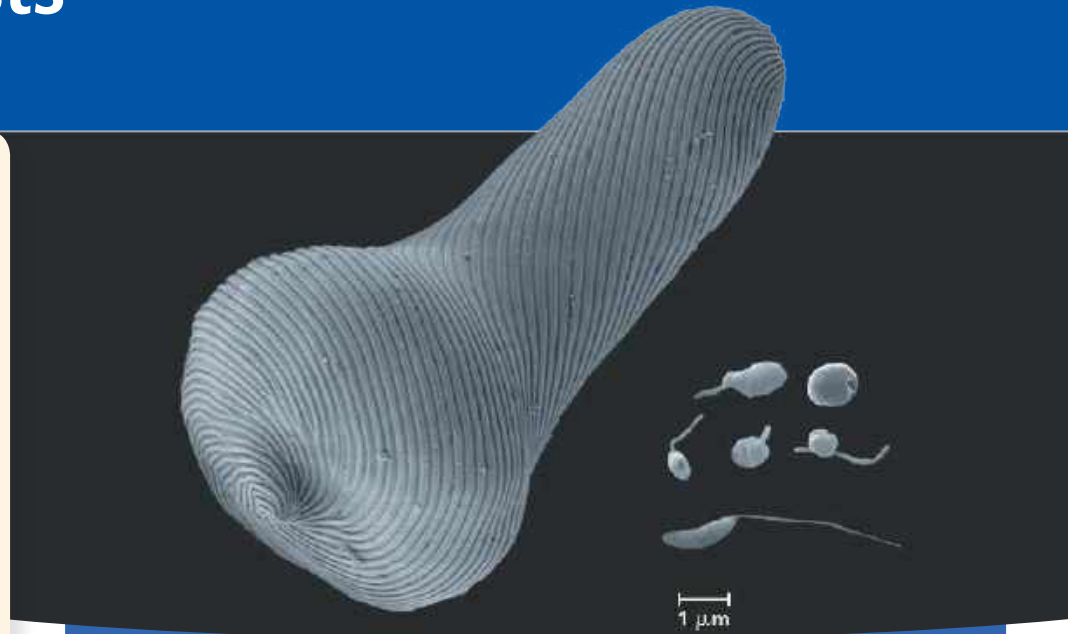
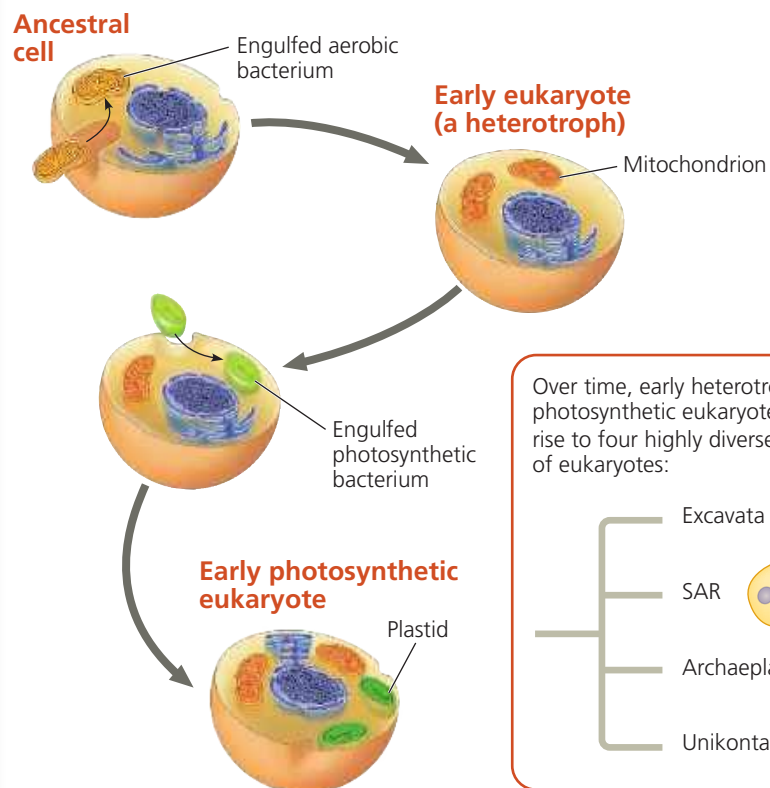


Figure 28.1 Knowing that prokaryotes are extremely small, you might assume that this image shows six prokaryotes and one much larger eukaryote. But in fact, the only prokaryote is the organism immediately above the scale bar—the rest are members of diverse, mostly unicellular groups of eukaryotes informally known as **protists**.

What gave rise to the great diversity of protists, and how have their lineages diverged over time?

Much of protistan diversity can be traced to **endosymbiosis**:



Over time, early heterotrophic and photosynthetic eukaryotes have given rise to four highly diverse **supergroups** of eukaryotes:

